

Non renormalizable theories

- can construct effective theories which are useful

22.1 Schrödinger eqn

consider

$$i\partial_t \psi = \left[-\frac{1}{2m} \nabla^2 + V(r) \right] \psi$$

- Non renormalizable
- Non relativistic
- Effective field theory

- p. unphysical / neg. mass dim's

→ write

$$H = \frac{\vec{p}^2}{2m} \left[1 + a_1 \frac{\vec{p}^2}{m} + a_2 \frac{\vec{p}^4}{m^2} + \dots \right] + V(r)$$

→ can obtain the renormalized Hamiltonian

$$(a_1 = -1/4, a_2 = -1/8, \dots)$$

$$E = \sqrt{\vec{p}^2 c^2 + m^2 c^4} \approx mc^2 + \frac{p^2}{2m} + \frac{p^4}{8m^3 c^2} + \dots$$

- Schrödinger eqn is predictive
w/ a $\alpha_1 = -1/4$ ($|\vec{p}| \ll m$ limit)

- Not predictive for

$$|\vec{p}| \sim m$$

- Dirac equation is a UV completion
of Schrödinger eqn

UV completion:

- Reduces to some theory at low energy
(like Schrödinger eqn)
but perturbatively breaks down at
high energy

22.24 - Fermi Theory

- Models weak (B) decay

$$\mathcal{L}_{\text{fermi}} = G_F \bar{\psi}_1 \psi_2 \bar{\psi}_3 \psi_4$$

(change 3 into 4)

- known as 4-fermion interaction
- successful model of radioactive decay

$$G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$$

$$= \left(\frac{1}{246.9 \text{ GeV}} \right)^2$$

- extracted from experiment

$$[G_F] = -2$$

→ not renormalizable

Counter terms:

$$\mathcal{L} = G_F \bar{\psi} \psi \bar{\psi} \psi + a_1 G_F^2 \bar{\psi} \psi \square \bar{\psi} \psi$$

$$+ a_2 G_F^3 \bar{\psi} \psi \square \bar{\psi} \psi$$

$$+ \dots$$

- representative; γ matrices can be inserted

- correction terms affect β rate by

$(G_F E^2)^2$ factors

- 4- Fermi can be used to study
parity violation

(compare $\bar{\psi}\psi \bar{\psi}\psi$ to $\bar{\psi}\gamma_5 \psi \bar{\psi}\gamma_5 \psi$)

Loops:

- suppose fermions identical
(for simplicity)

→ tree level for

$\psi\psi \rightarrow \psi\psi$

tree level:

$$M_{\text{tree}}(s) \sim G_F + a_1 G_F^2 s + a_2 G_F^3 s^2$$

- a is in any l.t. quantity of mass dim.

- low energies $\propto G_F^{-1}$, dominated
by leading term

- loops:

$$M_{\text{loop}}(s) = \text{diagram} \sim G_F^2 \int \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2} \frac{1}{k^2}$$

$$\sim G_F^2 \int b_0 \Delta^2 + b_1 s + b_2 s \ln \frac{\Delta^2}{s}$$

$$\therefore M_{\text{tree}} + M_{\text{loop}} \sim (G_F b_0 \Delta^2 G_F^2)$$

$$+ s G_F^2 (a_1 + b_1 + b_2 \ln \frac{\Delta^2}{s}) \\ - b_2 G_F^2 s \ln s + a_2 G_F^3 s^2 \ln s$$

(grouped by number of s)

key: s terms formed as counter part
in normal Feynman

Renormalize:

Add counter for g :

$$\mathcal{L} = \mathcal{L}_F \bar{\psi} \psi + \mathcal{L}_1 a_1^2 \bar{\psi} \psi \square \bar{\psi} \psi \\ + \mathcal{L}_2 a_2 G_F^3 \bar{\psi} \not{\partial} \psi \square \not{\partial} \psi \\ + \dots$$

$$\rightarrow Z_F = 1 + \delta_F, \quad Z_1 = 1 + \delta_1, \quad \dots$$

General comment:

"Non-analytic way dependence is a
measurable quantity per direction of
time re normalizable theories"

22.2.1) 4-fermion VV completion!

energy approach

- $A g^2 G_F^{-1/2} \sim 300 \text{ GeV}$
perturbative breaking

- VV completion includes
imaginary vector boson
with bosons & Z boson
(charged) (neutral)

- will see later in electroweak
- based on $SU(2) \times U(1)$

Lagrangian (vector boson W_μ)
mass term

$$\mathcal{L}_m = -\frac{1}{4} F_{\mu\nu}^2 + \frac{1}{2} M^2 W_\mu^2 + \frac{1}{4} (g^2 + g'^2) W_\mu^2$$

$$F_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu$$

Matrix element:

(toy model of single gauge boson & single fermion)

$$iM = \cancel{p_1} \cancel{p_2} \cancel{p_3} \cancel{p_4} \sim (ig)^2 \bar{u}_2 \not{\epsilon}_\mu \not{\epsilon}_\nu u_1 \frac{-i(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2})}{s - m^2} \times \bar{u}_3 \not{\epsilon}_\nu u_4$$

($\mathcal{L}(1)$ approx)

At low energy, approx by

$$iM = \cancel{p_1} \cancel{p_2} \cancel{p_3} \cancel{p_4} = -i \frac{g^2}{m^2} \bar{u}_2 \gamma_\mu u_1 \bar{u}_3 \gamma_\mu u_4$$

(- is just negative contribution to $\mathcal{L}$ since we begin propagator with small $\mathcal{L}$.)

- same as 4-fermi ($G_F = \frac{g^2}{m^2}$)

(actual expression is

$$G_F = \frac{\sqrt{2}}{g} \frac{g_w^2}{m_W^2}$$

22.3 Theory of mesons

- Yukawa
- chiral lagrangian
 - invariant under SU(2) isospin
 - problem & vector boson a_μ^i $\mathcal{L}_2 = (\rho_\mu^i)^2$

→ Pion (direct strong between proton & neutron) respect
 $U(1)$

$$\rightarrow \text{pions} = \pi^+, \pi^-, \pi^0$$

a complex interaction w/ σ & pions particles
 i.e.

$$U(1) = \exp \left[\frac{i}{F_\pi} \begin{pmatrix} \pi^0(x) & \sqrt{2} \pi^+(x) \\ \sqrt{2} \pi^-(x) & -\pi^0(x) \end{pmatrix} \right] = \exp \left[\frac{i}{F_\pi} \sigma_{\pi^0} \pi^0(x) \right]$$

$$F_\pi = \text{ion decay const.}$$

a simplest form of u

$$\mathcal{L}_\chi = \frac{F_\pi^2}{4} + u \left[(D_\mu u) (D_\mu u) \right]^\dagger$$

$$u / D_\mu = \partial_\mu - i g_\mu A_\mu$$

$$\rightarrow \mathcal{L}_{kin} = \frac{1}{2} (\partial_\mu \pi^0)^2 + (\partial_\mu \pi^\pm)^2$$

higher or low gives

$$\mathcal{L}_{int} = \frac{1}{F_\pi^2} \left[-\frac{1}{2} \pi^0 \pi^0 \partial_\mu \pi^+ \partial_\mu \pi^- + \dots \right]$$

$$\frac{1}{F_H} \left[\frac{1}{16} (\pi^\mu \pi^\mu)^2 \partial_\mu \pi^\nu \partial_\mu \pi^\nu \right]$$

→ not renormalizable
 $(\frac{1}{F_H})^2 = -1)$

- Add more terms:

$$L \sim L \sim 1$$

→ only 4 derivative terms that exist

$$L_4 = L_4 + \left[(\partial_\mu U) (\partial_\mu U)^\dagger \right]^2 \\ + L_2 + \left[(\partial_\mu U) (\partial_\mu U)^\dagger \right]^2 \\ + L_3 + \left[(\partial_\mu U) (\partial_\mu U)^\dagger (\partial_\nu U) (\partial_\nu U)^\dagger \right]$$

- quantum effects give non-analytic log. corrections
 → chiral logs

- non-perturbative at

$$\sqrt{s} \sim 4\pi F_\pi \sim \mu_{chiral}$$

- UV completion is QCD

- Quantum Gravity

- perturbative at $E_{\text{cutoff}} \sim 10^{16} \text{ GeV}$

22.2 Magnifying

- positive mass dimension never

Regular Manager:

$$- \text{consider } \mathcal{L} = -\frac{1}{2} \bar{\psi} (i \not{\partial} + m) \psi + \lambda \bar{\psi} \psi + \bar{\psi} (i \not{\partial} - \mu) \psi$$

$$+ \lambda d\psi + \psi (i\gamma - m)\psi$$

- effect of fermion loop $p = q$

Frühling 1892

$$\Sigma_2(\phi) = \frac{r+u}{r} \phi$$

$$= (i\lambda)^2 \int d^4x \frac{\gamma_\nu [(p + \not{x} + m)(\not{x} + m)]}{[p^2 + m^2 + i\epsilon][\hat{x}^2 - m^2 + i\epsilon]}$$

- Trail:

$$\text{Trace: } \text{Tr}[(A + iM)(A + M)] = 4(a^2 + b^2 + M^2)$$

9th fl 6th 9th - 1st (1-9) (drop from 1st in the in num.)

$$-1 \cdot \chi_2(r^2) = -q r^2 \int_0^{\pi} \int_0^{\pi} \frac{r^2 + p \cdot k + m^2}{[r^2 + (2r \cdot k + r^2)(1-q) - m^2 + ia]^2}$$

$$= -4\lambda^2 \int_0^1 dz \left(\frac{1}{k^2 - \Delta} + \frac{2\Delta}{(k^2 - \Delta)^2} \right)$$

$$\Delta = M^2 - p^2 u(1-u)$$

Dim Reg:

$$|\mathcal{L}_2(p^2)| = \frac{4\lambda^2(d+1)}{(4\pi)^{d/2}} \Gamma(1-d/2) M^{4-d} \int_0^1 dz (M^2 - p^2 u(1-u))^{d/2-1}$$

Expand $d=4-\epsilon$

$$\mathcal{L}_2(p^2) = -\frac{\lambda^2}{4\pi^2} \left\{ \frac{6M^2}{\epsilon} - \frac{p^2}{\epsilon} + M^2 - \frac{1}{6} p^2 + \int_0^1 dz \left(-p^2 u(1-u) - 3M^2 \right) \ln \frac{M^2 - p^2 u(1-u)}{4\pi M^2 e^{-\gamma_E}} \right\}$$

- div. prop to p^2 & M^2

- Dim Reg hides quadratic div. around $d=4$

Use different regulator:

$$\text{Derivative method (9-12e)} \\ \mathcal{L}_2(p^2) = \frac{3\lambda^2}{4\pi^2} \int_0^1 dz \left(M^2 - p^2 u(1-u) \right) \ln \frac{M^2 - p^2 u(1-u) + \Lambda^2}{\Lambda^2} \Big|_{\text{finite}}$$

- div. prop to p^2 & constant

div. prop to p^2 & M^2 (conceptually Dr. 19th century but more modern)

$$i(p^2 \delta q - (\delta m + \delta q) m^2_e)$$

w/ on shell renorm,
we L. equal to renorm mass (vrs 1)
pde

$$\therefore iG(p^2) = \frac{i}{p^2 - m^2 + \Sigma(p^2)}$$

$$w/ \Sigma(p^2) = \Sigma_c(p^2) + p^2 \delta_d - (\delta_m + \delta_d) m^2 p$$

- on shell conditions

$$G(m_p^2) = 0$$

$$\Sigma'(m_p^2) = 0$$

$$\text{at } m_p = m_p, \quad \delta_d = - \frac{2 \Sigma_c(p^2)}{\delta p^2} \Big|_{p^2 = m_p^2}$$

$$= (\delta_m = - \frac{2}{m_p} \Sigma_c(m_p^2)), \quad \delta_d = - \frac{2 \Sigma_c(p^2)}{\delta p^2} \Big|_{p^2 = m_p^2}$$

- arising calculation for $\delta \phi$

$$\delta_m = - \frac{\lambda^2}{4\pi^2} \left[\left(\frac{6m^2}{m_p^2} - 1 \right) \frac{1}{\epsilon} + \left(\frac{1}{\epsilon} - \frac{3m^2}{m_p^2} \right) \ln \frac{m^2}{m_p^2} + \frac{\mu^2}{m_p^2} + \frac{1}{3} \right. \\ \left. - \frac{m^2 p}{20m^2} + \mathcal{O}\left(\frac{m^4}{m^6}\right) \right]$$

$$\delta_d = - \frac{\lambda^2}{4\pi^2} \left[\frac{2}{\epsilon} - \frac{1}{\epsilon} \frac{m m^2}{m^2} - \frac{1}{3} + \frac{m^2}{10m^2} + \mathcal{O}\left(\frac{m^4}{m^6}\right) \right]$$

for $p^2, m_p^2 \ll m^2$,

$$\Sigma(p^2) = \Sigma_c(p^2) + p^2 \delta_d - (\delta_m + \delta_d) m^2 p = \frac{\lambda^2}{4\pi^2} \left[\frac{(p^2 - m_p^2)^2}{20m^2} + \mathcal{O}\left(\frac{p^4}{m^4}\right) \right]$$

- finite

- sometimes more efficient to use MS than on shell

- IRs counterterms:

$$\delta q = \frac{-\lambda^2}{4\pi^2 \epsilon}, \quad \delta m = \frac{-\lambda^2}{4\pi^2 \epsilon} \left(\frac{6m^2}{m_p^2} - 1 \right)$$

pole mass diff.:

$$m_p^2 - m_{\text{ph}}^2(m)$$

$$= \frac{\lambda^2}{24\pi^2} (6m^2 - m_p^2) - \frac{3\lambda^2}{4\pi^2} \int_0^1 dx [M^2 - m_p^2 x(1-x)] \ln \frac{M^2 - m_p^2 x(1-x)}{m^2}$$

- finite

- $m \rightarrow \infty \rightarrow$ very large

- UV sensitive

22. $\phi \rightarrow$ Fermion & Gauge masses

As calculated in QED,

$$i\Sigma_2(p) = \text{diagram: fermion line with a photon loop}$$

$$= -i \frac{e^2}{8\pi^2} \int_0^1 dx (2m_p - xq) \left(\frac{2}{\epsilon} + \ln \frac{m^2}{(1-x)(m_p^2 - q^2 x) + x m_p^2} \right)$$

$\rightarrow$ diff

$$m_p - m_{\text{ph}} = \frac{e^2}{6\pi^2} m_p \left(5 + 3 \ln \frac{m^2}{m_p^2} \right)$$

- No pole in $d=2$ or $d=3$

$\rightarrow$ w/ quant. or kin. dep.

- Neutral

- correction prop. to only m_p

- if A is replaced w/ other mass like $m \rightarrow$ still prop. to m_p
 generalizes to Yukawa theory

$$i\Sigma_2(p) = \text{diagram: fermion line with a scalar loop}$$

$$= \lambda^2 \int d^4k \frac{k^2 - k + m}{[(k-p)^2 - m^2][k^2 - m^2]}$$

$$= i \frac{\int \frac{d^4 p}{(2\pi)^4} \left[\frac{p^2 + 2M}{i} - \int_0^1 dx [(1-x)(p^2 + M^2)] \frac{M^2 x + (1-x)(m^2 - p^2 x)}{p^2} \right]$$

- only correction prop. to M

- w/ bare quarks, not prop/c. fermion mass

- This is b.c. m_e protected by exact chiral symmetry

(chiral symmetry:

- left & right handed electron have opposite charge

$$\psi_L \rightarrow e^{-i\alpha} \psi_L, \quad \psi_R \rightarrow e^{i\alpha} \psi_R$$

(conjugate:

$$\psi \rightarrow e^{i\alpha} \psi$$

in lag, kinetic term:

$$\psi \not{\partial} \psi \rightarrow \psi \not{\partial} e^{-i\alpha} \psi, \quad \psi \not{\partial} e^{i\alpha} \psi = \psi \not{\partial} \psi$$

$$\text{since } \not{\partial}_5^\dagger = \not{\partial}_5, [\not{\partial}_5, \not{\partial}_\mu \not{\partial}_\mu] = 0$$

mass term not

$$m_e \bar{\psi} \psi \rightarrow m_e \bar{\psi} e^{i\alpha} \psi \neq m_e \bar{\psi} \psi$$

$\rightarrow$ mass term breaks symmetry

consistent w/ mass approx:

$$\bar{\psi} \not{\partial} \psi + m_e \bar{\psi} \psi = \bar{\psi}_R \not{\partial} \psi_R + \bar{\psi}_L \not{\partial} \psi_L + \underbrace{m_e \bar{\psi}_L \psi_R + m_e \bar{\psi}_R \psi_L}_{\text{L & R comp/c}}$$

$\rightarrow$ mass couples L & R

(chiral sym is $\lim_{m_e \rightarrow 0}$)

$\rightarrow$ if $m_e = 0$, will only 0 & codim of partition theory

- Custodial symmetry prevents mass from being corrected

- no mass for

$$L = \dots + \bar{\psi} \not{\partial} \psi$$

→ gauge invar. needs to be broken!

21.7 Superrenormalizable theories

- only from in loop w/ mass dim 4:

$$L = \dots + \mathcal{O}$$

$$dim[\mathcal{O}] = 4$$

ρ is cosmological const.

only useful in gravity

$$expanding g_{\mu\nu} = \eta_{\mu\nu} + \frac{1}{M_{Pl}^2} h_{\mu\nu}$$

$$\rightarrow L = \sqrt{g}(\mathcal{L}(\rho)) = \frac{1}{2} \frac{1}{M_{Pl}^2} h_{\mu\nu} \rho + \frac{1}{M_{Pl}^2} \bar{h}_{\mu\nu}^2 \rho + \dots + \sqrt{g} \mathcal{R}$$

- treat term given above $\mathcal{L}(\rho) / h_{\mu\nu}(\rho)$

→ expanding about many masses

→ define

$$h_{\mu\nu} \rightarrow h_{\mu\nu}^c + h_{\mu\nu}$$

w/ gauge $h_{\mu\nu}^c(x)$ & $\mathcal{R}[h_{\mu\nu}^c] = \rho$
 generalized
 - cosmological const. small

$$\rho \sim 10^{122} / M_{Pl}^4$$

- times w/ $\mathcal{L}(\rho)$ → cause tad poles

2.2.2 relevant Interactions

- consider $g \phi^3$

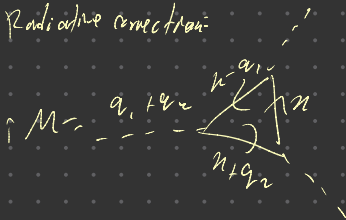
$$\rightarrow \mathcal{L} = -\frac{1}{2} (\Box + m^2) \phi + \frac{g}{3!} \phi^3$$

- consider $\rightarrow$ propagator function

tree level,

$$M(p, q_1, q_2) = \text{---} \text{---} \text{---} = -g$$

Radiative correction:



$$= -g^3 \int \frac{d^4 k}{(k^2 - m^2) [(k - q_1)^2 - m^2] [(k + q_2)^2 - m^2]}$$

- obviously UV finite

- magnetic. of IR l.u.

consider low energy (long dist.)

$E \ll m$

$$\rightarrow M \sim \frac{g^3}{m^2} (\dots)$$

higher order

$$M \sim g \left(1 + \frac{\theta^2}{m^2} + \left(\frac{\theta^2}{m^2} \right)^2 \dots \right)$$

in $\mathcal{L}_g$, perturbation $\sim \text{useful}$

Also: $m^2 \rightarrow \nu \rightarrow \bar{g}^2$, perturbation breaks

→ general case per nonrenormalizable
heavy feature; breaks all
high d. low dist.